BUSM 410 Semester Project — Stage 1: Discovery, Problem Framing & Empirical Gap Charter

Project Focus: NYC Citi Bike Asset Utilization & Station Performance
Platform: Google Cloud BigQuery Studio (`bigquery-public-data.new_york_citibike`)
Status: Stage 1 Complete | Ready for Stage 2 Benchmark Launch

1. Executive Summary & Problem Statement

Citi Bike operates a large network of physical stations across New York City. Each station consists of physical docking slots that carry fixed installation, maintenance, and public space footprints.

Using BigQuery to analyze the full historical trip dataset (`58,937,715` trips) and focusing on a mature, weather-stable representative month (**September 2017**, 541 active stations), we measured **Station Dock Turnover Rate** (Daily Trip Starts/Dock Capacity
Daily Trip Starts/Dock Capacity).

Key Empirical Findings

- **System Median Benchmark: 1.59 rides / dock / day**.
- **Underutilized Cohort (Bottom Quartile): 135 stations** (25% of the network) average only **0.58 rides / dock / day**.
- **The Operational Gap:** A deficit of **1.01 rides / dock / day** (**63.4% shortfall** below the system median).
- **Physical Assets Trapped:** These 135 underperforming stations tie up **4,675 physical docks**.
- **Missed Opportunity:** If this bottom quartile achieved the system median turnover, Citi Bike would have generated **4,716 additional rides per day**, or **~141,490 unrealized rides in September 2017 alone**.

2. Table Inspection & Schema Architecture

Verified Tables in `bigquery-public-data.new_york_citibike`

Table Name	Verified Rows	Date Coverage	Key Fields Used
`citibike_trips`	**58,937,715**	`2013-07-01` to `2018-05-31`	`starttime` (TIMESTAMP), `start_station_name` (STRING)
`citibike_stations`	Static Snapshot	N/A (Station Metadata)	`name` (STRING), `capacity` (INT64), `latitude`, `longitude`

Critical Schema Quirk Resolved

- **Mismatch Identified:** In `citibike_stations`, `station_id` contains UUID strings (e.g. `'66dc292c-0aca-11e7-82f6-3863bb44...'`) and `short_name` contains modern DOT decimal codes (`"6560.01"`), while `citibike_trips.start_station_id` contains legacy integers.
- **Resolution:** Tables must be joined on station name:

  ```sql
  ON t.start_station_name = s.name
  ```

- **Query Scan Impact:** Because `citibike_trips` is unpartitioned in the public dataset, scanning `starttime` (8 bytes) + `start_station_name` (~25 bytes) processes **~2.57 GB**. Running this once produces a compact 541-row summary table that costs **< 1 MB** for all subsequent analysis.

3. Final Stage 1 Gap Charter

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BUSM 410 STAGE 1 GAP CHARTER: BIKE-SHARE DOCK ASSET UTILIZATION
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1. Management Question:
   Which Citi Bike stations chronically underperform the network median
   turnover benchmark, and what station characteristics are associated with
   this 63.4% operational utilization deficit?

2. KPI Definition and Unit:
   - Metric: Station Dock Turnover Rate
   - Operational Formula: (Daily Trip Starts) / (Station Dock Capacity)
   - Unit: Rides per Dock per Day (trips/dock/day)

3. Population & Time Period:
   - Population: 541 active Citi Bike stations in NYC.
   - Time Period: September 1, 2017 – September 30, 2017 (30 calendar days).
   - Historical Context: Part of the verified 58.9M trip population (2013–2018).

4. Business Gap / Missed Opportunity (Empirically Measured):
   - 135 stations (25% of the network) operate at an average turnover of only
     0.58 rides/dock/day compared to the network median of 1.59.
   - Gap Size: A deficit of 1.01 rides/dock/day (63.4% below benchmark).

5. Benchmark and Rationale:
   - Benchmark: 1.59 rides/dock/day (System Median Station Turnover).
   - Rationale: The median represents typical operational network performance,
     controlling for macro weather and seasonal conditions in September 2017
     without being skewed by extreme outliers (e.g., Great Jones St at 12.59).

6. Verified Required Fields:
   - `citibike_trips`: `starttime` (TIMESTAMP), `start_station_name` (STRING)
   - `citibike_stations`: `name` (STRING), `capacity` (INT64), `latitude`,
`longitude`

7. Quantified Business Value:
   - Operational Value: 4,716 unrealized daily trips across 4,675 underperforming
     docks, totaling ~141,490 unrealized rides in September 2017.
   - Grounded strictly in physical asset efficiency and ride volume to avoid
     speculative dollar/pricing assumptions.

8. Important Assumptions and Limitations:
   - Station capacity is treated as static across September 2017.
   - Measures realized demand (completed trips), not unobserved lost demand from
     empty docks.
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   - Statistical associations do not prove causation.
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```

4. Empirical Distribution Data (September 2017)

Across 541 active stations matched between `citibike_trips` and `citibike_stations`:

Metric	Turnover (Rides/Dock/Day)	Daily Starts	Capacity (Docks)	Example Station
Minimum	**0.07**	1.3	19	`34 Ave & 13 St` (Queens)
25th Percentile (Q1)	**0.99**	23.8	24	`10 St & 5 Ave`
Median (Benchmark)	**1.59**	58.3	39	`Carroll St & Bond St`
Mean	**1.98**	86.4	43	System Average
75th Percentile (Q3)	**2.68**	146.6	54	`E 48 St & 5 Ave`
Maximum	**12.59**	289.5	23	`Great Jones St` (NoHo)

5. Verified BigQuery SQL Queries

Query 1: Stage 1 Dataset Boundary Validation

```sql
-- Validates total row count and date range in citibike_trips
-- Scanned: ~471 MB
SELECT
  MIN(starttime) AS earliest_trip,
  MAX(starttime) AS latest_trip,
  COUNT(1) AS total_trips
FROM
  `bigquery-public-data.new_york_citibike.citibike_trips`;
```

Output:

- `earliest_trip`: `2013-07-01 00:00:00 UTC`
- `latest_trip`: `2018-05-31 23:59:59.606 UTC`
- `total_trips`: `58,937,715`

Query 2: September 2017 Station Turnover & Benchmark Measurement

```sql
-- Computes 30-day average daily turnover per station joined on name
-- Scanned: ~2.57 GB
WITH station_daily_trips AS (
  SELECT
    DATE(t.starttime) AS trip_date,
    t.start_station_name,
    COUNT(1) AS daily_starts
  FROM
    `bigquery-public-data.new_york_citibike.citibike_trips` t
  WHERE
    t.starttime >= '2017-09-01' AND t.starttime < '2017-10-01'
    AND t.start_station_name IS NOT NULL
  GROUP BY
    trip_date, start_station_name
),
station_daily_turnover AS (
  SELECT
    d.trip_date,
    d.start_station_name,
    d.daily_starts,
    s.capacity,
    SAFE_DIVIDE(d.daily_starts, s.capacity) AS dock_turnover
  FROM
    station_daily_trips d
  INNER JOIN
    `bigquery-public-data.new_york_citibike.citibike_stations` s
    ON d.start_station_name = s.name
  WHERE
    s.capacity > 0
)
SELECT
  start_station_name,
  ROUND(AVG(dock_turnover), 2) AS avg_daily_turnover,
  ROUND(AVG(daily_starts), 1) AS avg_daily_starts,
  MAX(capacity) AS dock_capacity
FROM
  station_daily_turnover
GROUP BY
  start_station_name;
```

6. Stage 2 Benchmark Launch Plan (Prepared for Next Phase)

1. Recommended Benchmark Comparison

- **Station-to-Median Gap Analysis:** Measure every station relative to the **1.59 system median**, categorizing them into:
    - **High Demand (Top 25%):** > 2.68> 2.68 rides/dock/day
    - **Core Performers (Middle 50%):** $0.99 – 2.68$0.99 **–** 2.68 rides/dock/day
    - **Underutilized (Bottom 25%):** < 0.99< 0.99 rides/dock/day

2. Next Stage Benchmark Query (Ready to Execute)

```
sql
-- Stage 2: Generates station-level gap analysis with geographic coordinates
WITH station_daily_trips AS (
  SELECT
    DATE(t.starttime) AS trip_date,
    t.start_station_name,
    COUNT(1) AS daily_starts
  FROM
    `bigquery-public-data.new_york_citibike.citibike_trips` t
  WHERE
    t.starttime >= '2017-09-01' AND t.starttime < '2017-10-01'
    AND t.start_station_name IS NOT NULL
  GROUP BY
    trip_date, start_station_name
),
station_turnover AS (
  SELECT
    d.trip_date,
    d.start_station_name,
    d.daily_starts,
    s.capacity,
    s.latitude,
    s.longitude,
    SAFE_DIVIDE(d.daily_starts, s.capacity) AS dock_turnover
  FROM
    station_daily_trips d
  INNER JOIN
    `bigquery-public-data.new_york_citibike.citibike_stations` s
    ON d.start_station_name = s.name
  WHERE
    s.capacity > 0
)
SELECT
  start_station_name,
  ROUND(AVG(dock_turnover), 2) AS avg_daily_turnover,
  ROUND(AVG(daily_starts), 1) AS avg_daily_starts,
  MAX(capacity) AS dock_capacity,
  ANY_VALUE(latitude) AS latitude,
  ANY_VALUE(longitude) AS longitude,
  ROUND(AVG(dock_turnover) - 1.59, 2) AS gap_vs_median_turnover,
  ROUND(GREATEST(0, (1.59 - AVG(dock_turnover)) * MAX(capacity)), 1) AS
unrealized_daily_starts,
  CASE
```

```
    WHEN AVG(dock_turnover) < 0.99 THEN 'Underutilized (Bottom 25%)'
    WHEN AVG(dock_turnover) <= 2.68 THEN 'Core Performer (Middle 50%)'
    ELSE 'High Demand (Top 25%)'
  END AS performance_tier
FROM
  station_turnover
GROUP BY
  start_station_name
ORDER BY
  avg_daily_turnover ASC;
```

3. Chart Concepts for Presentation & Poster

1. **Chart 1 (The Benchmark Gap Distribution):** Histogram of 541 stations with a benchmark line at **1.59** and red shading for the bottom 25% cohort (< 0.99< 0.99).
2. **Chart 2 (Geographic Clustering Map):** NYC scatter plot mapping `latitude` and `longitude` colored by `performance_tier` to show where underutilized stations are geographically concentrated.